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BIOLOGY

0610/43

Paper 4 Theory (Extended)

May/June 2026

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [].

This document has **24** pages. Any blank pages are indicated.

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1 (a) Figure 1.1 shows the five kingdoms of living organisms.

(i) Complete Figure 1.1 by naming the missing kingdoms.

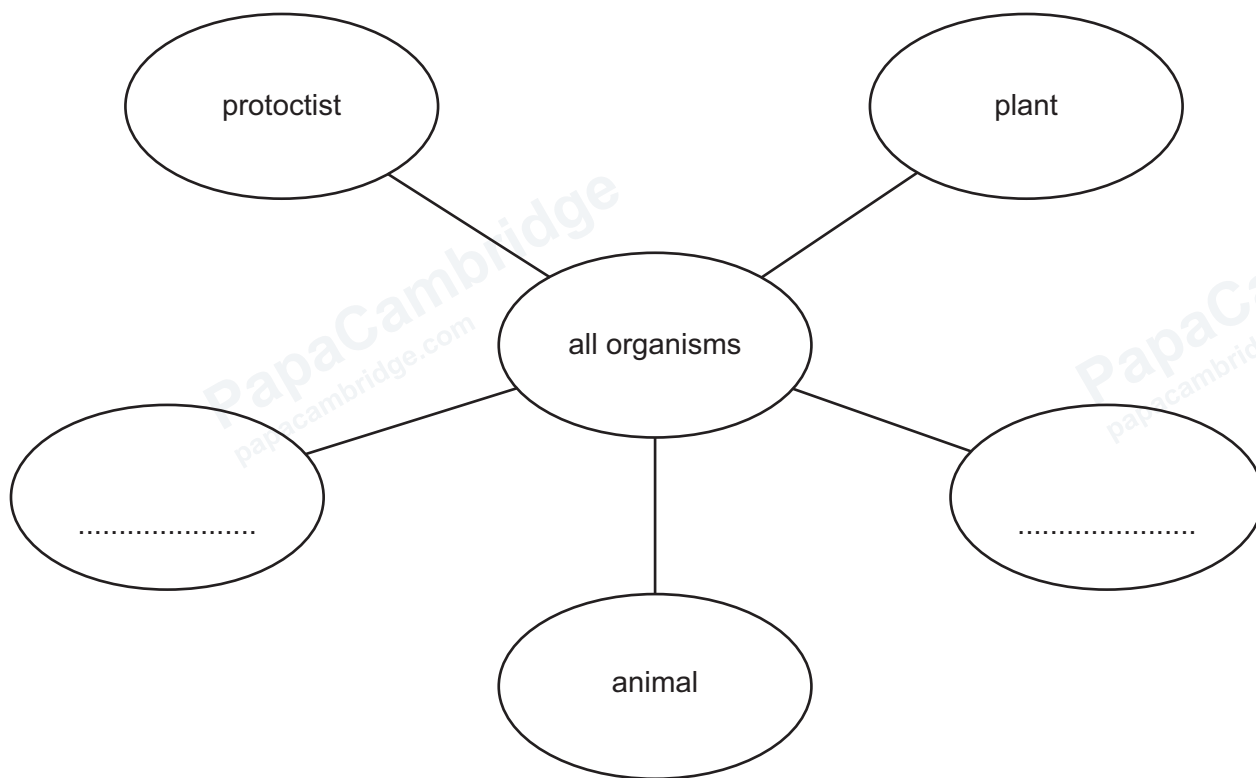


Figure 1.1

[2]

(ii) State the names of **two** cell structures that are found in both plant and animal cells.

1

2

[2]

(b) Figure 1.2 is a diagram of a single-celled protist called *Euglena*.

Euglena can act as a producer and as a consumer.

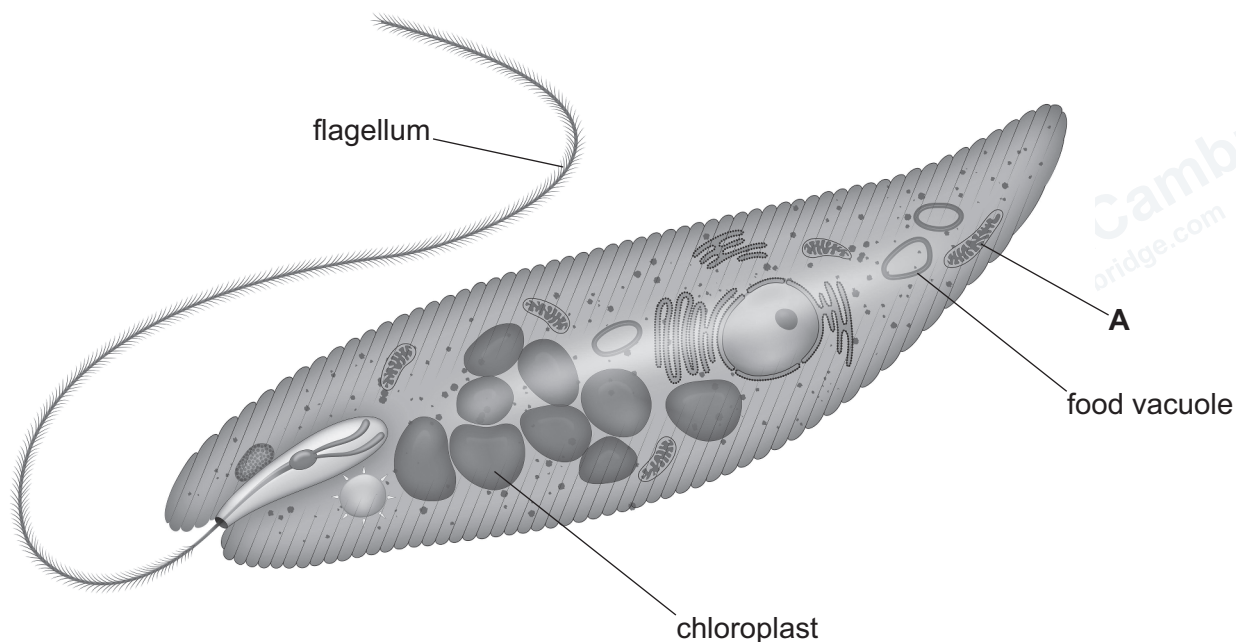


Figure 1.2

(i) State the evidence shown in Figure 1.2, that *Euglena* are producers.

.....

.....

..... [1]

(ii) Suggest the advantages to this organism of being a producer and a consumer.

.....

.....

.....

.....

..... [2]

(iii) State the function of the structure labelled **A** in Figure 1.2.

.....

..... [1]

- (iv) The prototist in Figure 1.2 takes up magnesium ions from its surroundings.

Suggest **one** use of the magnesium ions.

..... [1]

- (v) The prototist in Figure 1.2 has specialised structures that detect light.

Suggest which characteristic of living organisms is provided by the structures that detect light.

..... [1]

- (c) State the name of a human cell that has a flagellum **and** describe the function of the flagellum.

human cell

function

.....

[2]

[Total: 12]

- 2 (a) Metabolism produces waste substances that are excreted.

Complete Table 2.1 about the excretion of substances.

Table 2.1

substance	process that produces the substance	organ that excretes the substance
carbon dioxide		
.....	photosynthesis	

[2]

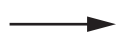
(b) Figure 2.1 shows the movement of nitrogen-containing substances through the human body.

key

substance



organ



direction of movement

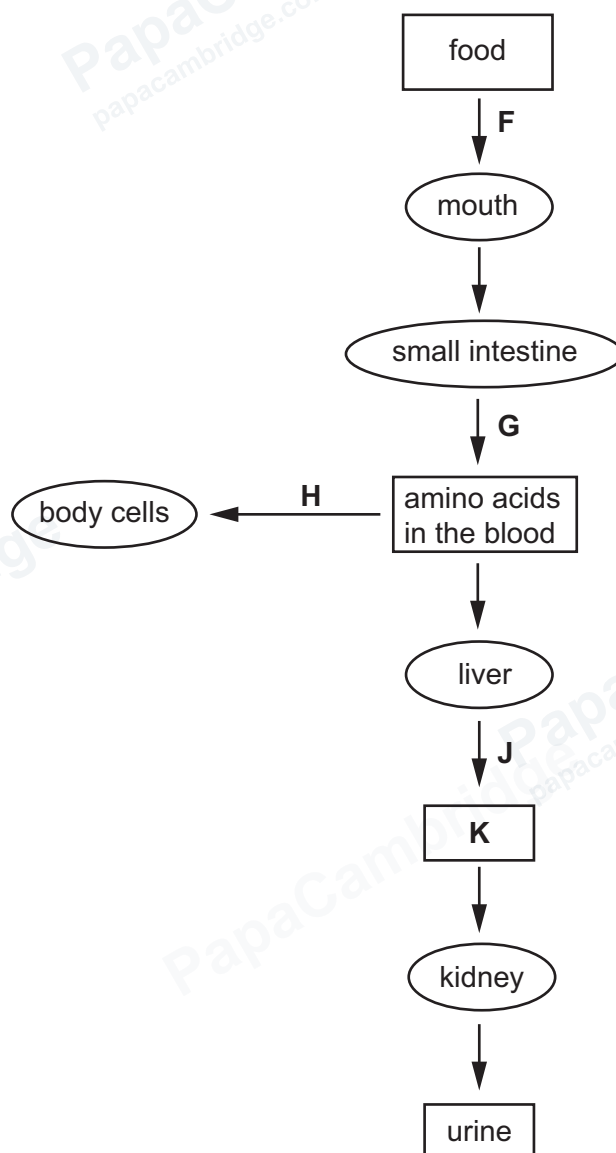


Figure 2.1



(i) Identify the processes occurring at **F**, **G**, **H** and **J** in Figure 2.1.

F

G

H

J

[4]

(ii) State the name of substance **K** in Figure 2.1.

..... [1]



(c) Figure 2.2 is a diagram of a nephron from a human kidney and its associated blood vessels.

The arrows show the direction of blood flow.

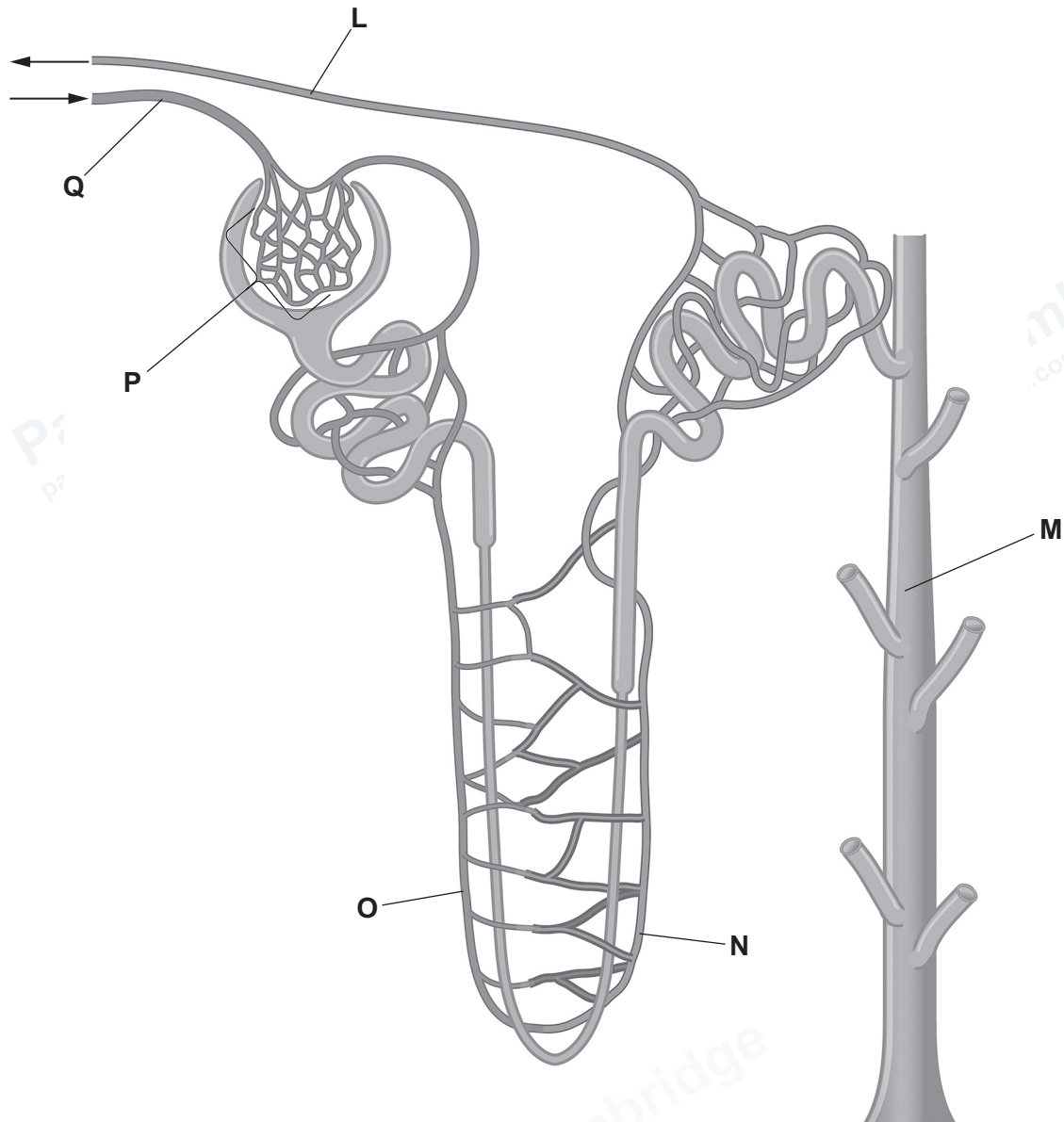


Figure 2.2

- (i) State the letter in Figure 2.2 that identifies the glomerulus **and** describe its function.

letter

function

[2]

- (ii) The oxygen concentrations of the blood in blood vessel **Q** and blood vessel **L** are different.

State **and** explain the difference in oxygen concentration between blood vessel **Q** and blood vessel **L**.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 13]

- (c) If pregnancy occurs, a placenta will form.

Describe **two** functions of the placenta.

1

.....

2

.....

[2]

- (d) FSH and LH are two of the hormones that regulate the menstrual cycle.

State the roles of FSH and LH.

FSH

.....

LH

.....

[2]

- (e) State **two** effects of testosterone during puberty in human males.

1

.....

2

.....

[2]

- (f) State the name of the component of blood that transports hormones.

..... [1]

[Total: 16]

- 4 (a) Figure 4.1 shows a cross between a chicken that is pure-breeding for white feathers and a chicken that is pure-breeding for black feathers. All of the offspring have black and white feathers that are described as speckled.

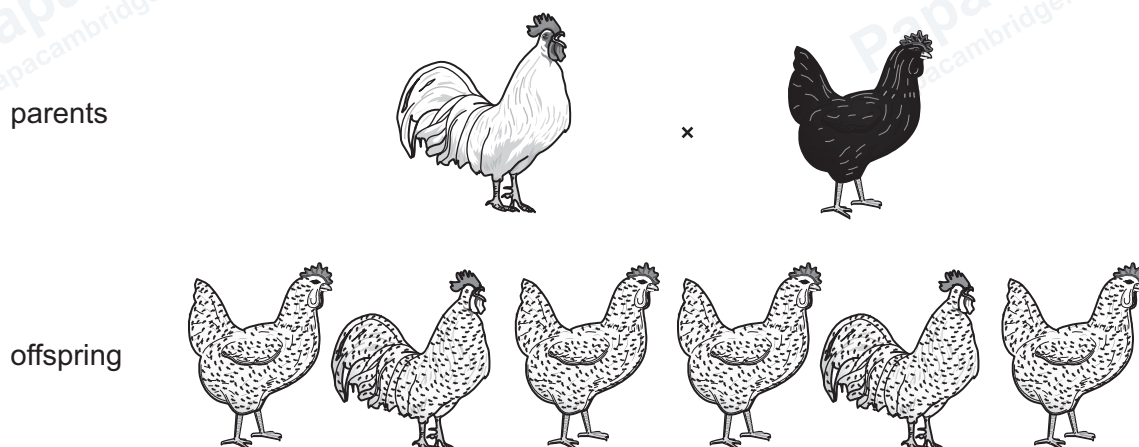


Figure 4.1

The allele for white feathers is represented by F^W .

The allele for black feathers is represented by F^B .

The genotype of all of the speckled chicken offspring is $F^W F^B$.

- (i) State the term that is used to describe the situation where both alleles in heterozygous organisms contribute to the phenotype.

..... [1]

- (ii) Two of the speckled chicken offspring were bred together.

Complete the Punnett square to determine the genotypes and phenotypic ratio of the offspring from this cross.

		male speckled chicken	
female speckled chicken			
			
female speckled chicken			
			

The phenotypic ratio of the offspring is:

..... [3]

- (b) Chickens can be farmed intensively.

Describe **two** disadvantages of intensive livestock production.

- 1
-
- 2
- [2]



Explain how a protein is made.

[6]

[Total: 12]



Turn over



5 Figure 5.1 shows a food web from a marine ecosystem.

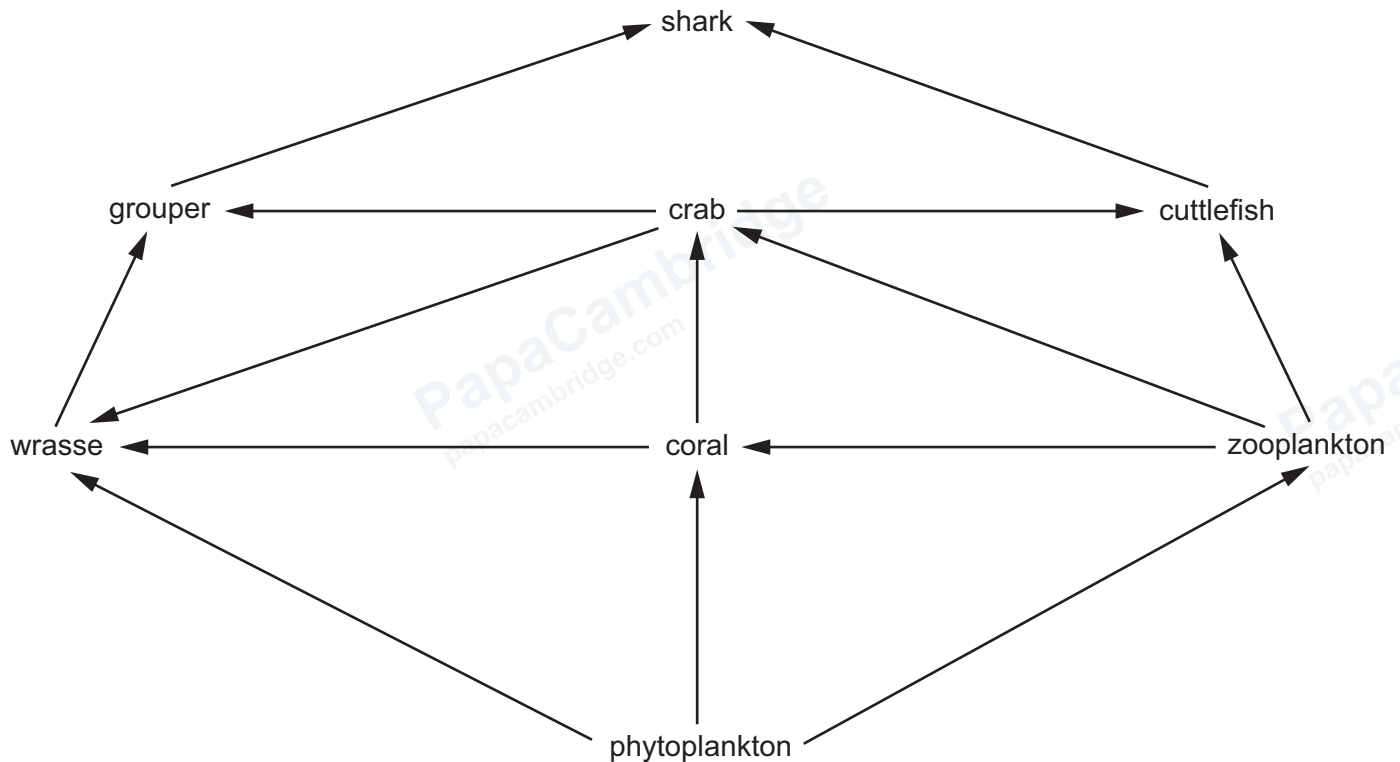


Figure 5.1

(a) Using Figure 5.1, identify:

an organism that **only** feeds as a secondary consumer

.....

an organism that consumes at three different trophic levels

.....

all of the quaternary consumers.

.....

[3]

(b) Figure 5.2 shows a food chain.

phytoplankton → zooplankton → cuttlefish

Figure 5.2

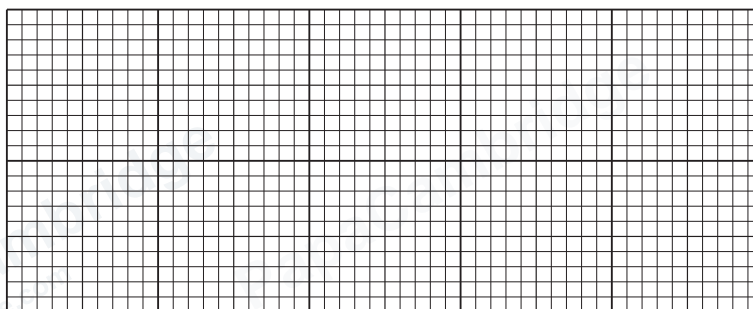
Table 5.1 shows the energy in the organisms shown in Figure 5.2.

Table 5.1

organism	energy/arbitrary units
phytoplankton	960
zooplankton	120
cuttlefish	10

- (i) Use the information in Figure 5.2 and Table 5.1 to construct a pyramid of energy on the grid.

Label each trophic level.



[3]

- (ii) State **one** reason for using a pyramid of energy rather than a pyramid of biomass.

.....

.....

..... [1]

(c) Phytoplankton are aquatic producers.

Scientists investigated the factors that affect their rate of population growth. A sample of phytoplankton was grown in a container with conditions similar to those found in the wild.

The phytoplankton were provided with constant light.

The population of phytoplankton was measured as the number of organisms per cm^3 .

Figure 5.3 shows the results of the investigation.

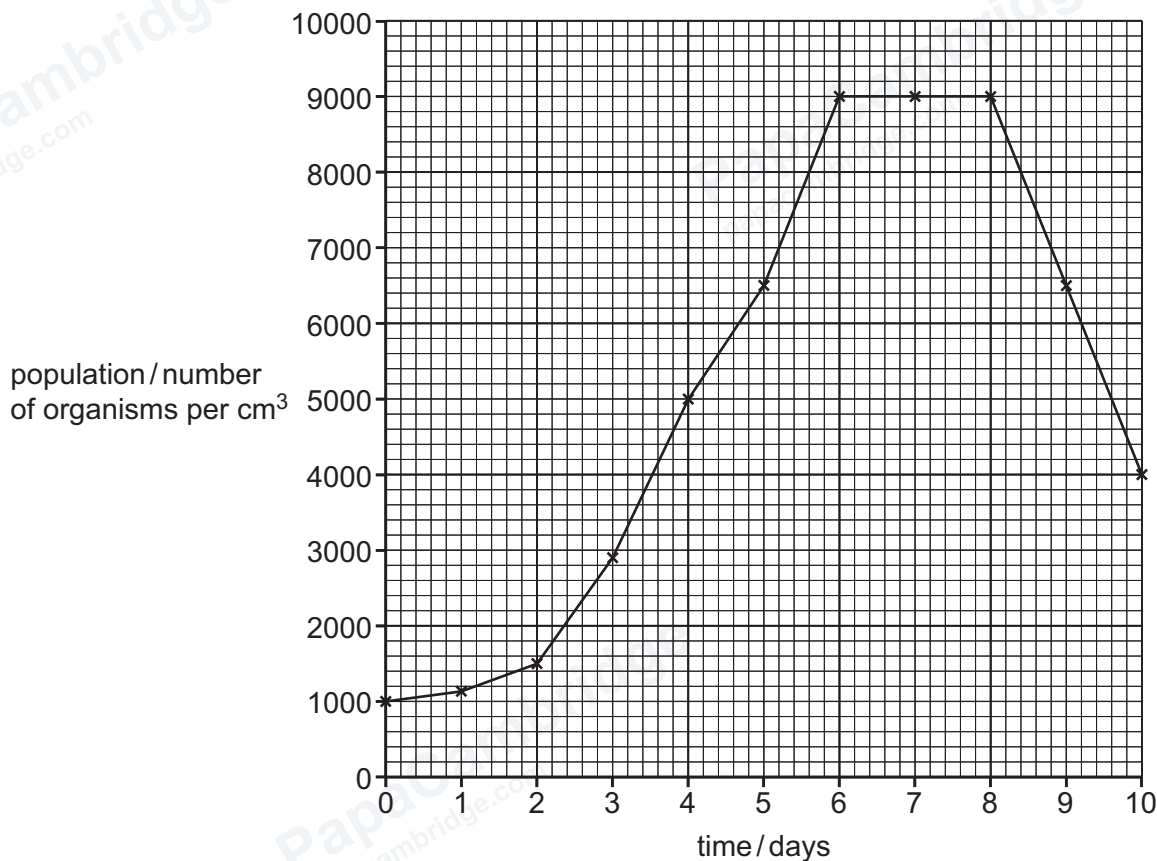


Figure 5.3

(i) Using Figure 5.3, calculate the population of phytoplankton in 0.5 dm^3 on day 4.

Give your answer in standard form.

Space for working.

..... [3]



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[5]

[Total: 15]



- 6 (a) Lipase is an enzyme found in the human digestive system.

Complete the word equation to show the substrate and products for a reaction catalysed by lipase.

..... → fatty acids +

[2]

- (b) There are different types of lipase.

Figure 6.1 shows the effect of pH on the enzyme activity of two types of lipase, type **P** and type **Q**.

key

— type **P** lipase

--- type **Q** lipase

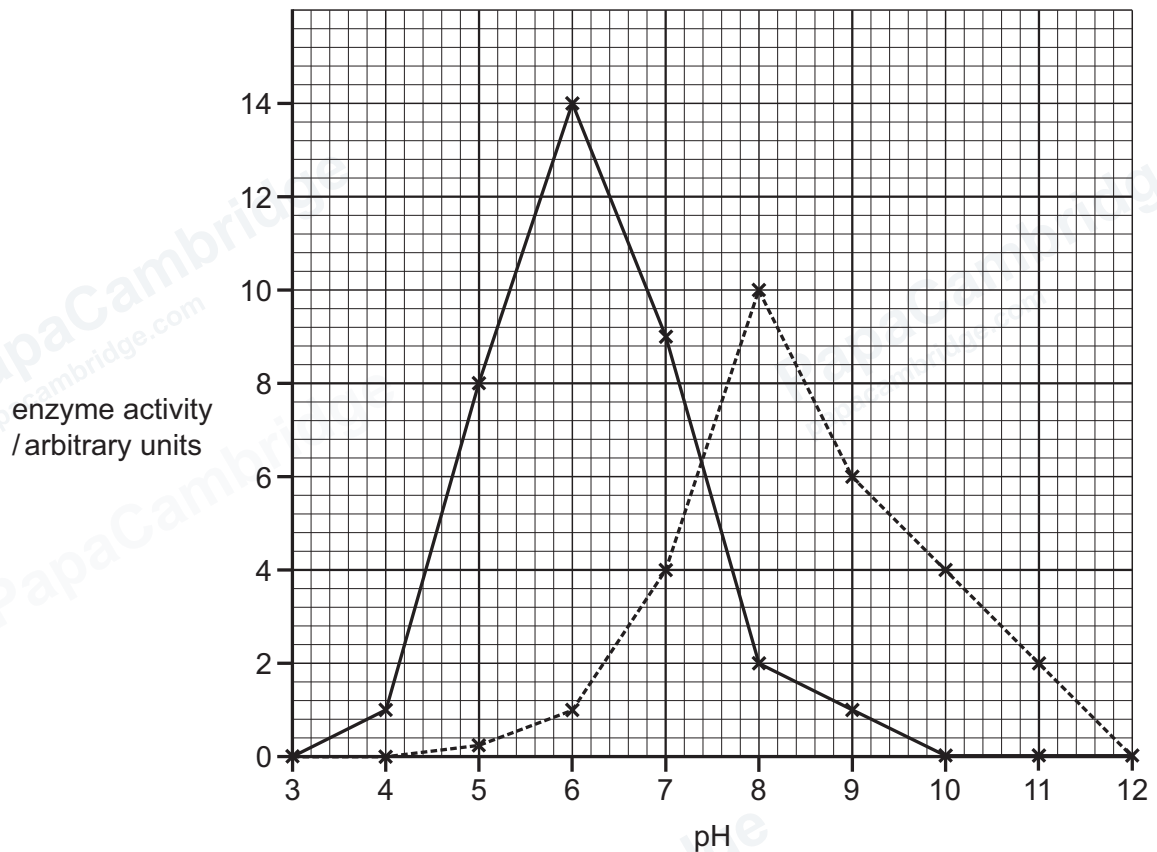


Figure 6.1

- (i) Describe **two** similarities and **two** differences between the effect of pH on the enzyme activity of type **P** lipase and type **Q** lipase.

similarity 1

.....

similarity 2

.....

difference 1

.....

difference 2

.....

[4]

- (ii) Using Figure 6.1, identify the pH values where type **Q** lipase has 50% of its maximum activity.

..... and [1]

- (iii) Explain the effect of pH on the enzyme activity of type **P** lipase shown in Figure 6.1.

.....

.....

.....

.....

.....

.....

..... [3]

- (c) In the small intestine, bile provides a suitable pH for type **Q** lipase.

- (i) State **one** other function of bile.

.....

..... [1]

- (ii) State the name of the organ that produces bile.

..... [1]

[Total: 12]







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